

```
install.packages("mlbench")
```

```
↗ Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
library(mlbench)
library(dplyr)
```

```
data("BostonHousing2")
dat <- BostonHousing2 %>%
  select(-c("town", "lat", "lon", "tract", "medv"))
```

```
fit.full <- lm(cmedv ~ ., data = dat)
```

```
fit.null <- lm(cmedv ~ 1, data = dat)
```

```
fit.forward <- step(fit.null,
  scope = list(lower = ~1, upper = fit.full),
  direction = "forward",
  trace = FALSE,
  k = 2,
  keep = function(model, aic) list(model = model, aic = aic))
```

```
fifth_model <- fit.forward$keep[[5]]$model
print(summary(fifth_model))
```

```
↗
```

	cmedv	lstat	rm
Min.	: 5.00	Min. : 1.73	Min. :3.561
1st Qu.:	17.02	1st Qu.: 6.95	1st Qu.:5.886
Median :	21.20	Median :11.36	Median :6.208
Mean :	22.53	Mean :12.65	Mean :6.285
3rd Qu.:	25.00	3rd Qu.:16.95	3rd Qu.:6.623
Max.	:50.00	Max. :37.97	Max. :8.780

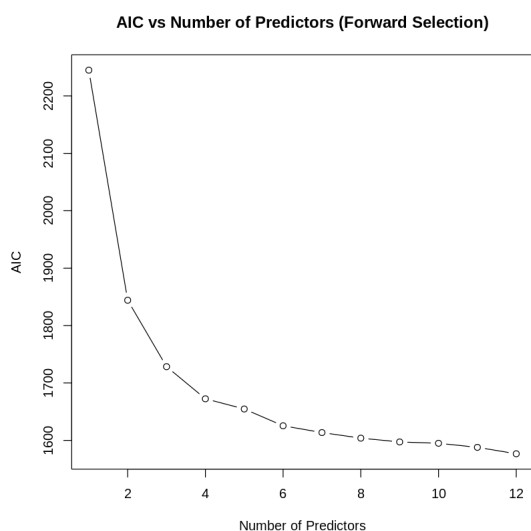
```
aic_values <- sapply(fit.forward$keep, function(x) {
  if (is.numeric(x)) return(x)
  return(NA)
})
```

```
aic_values <- na.omit(aic_values)
```

```
num_predictors <- seq_along(aic_values)
```

```
plot(
  num_predictors, aic_values, type = "b",
  xlab = "Number of Predictors", ylab = "AIC",
  main = "AIC vs Number of Predictors (Forward Selection)"
)
```

```
↗
```



```
print(fit.forward$keep)
```

```

[1,]      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]      [,8]
model lm,12      lm,12      lm,12      lm,12      lm,12      lm,12      lm,13      lm,13
aic    2244.87    1844.13    1728.426  1672.444  1654.666  1625.411  1613.595  1603.882
bic    3691.289    3294.776    3183.298  3131.542  3117.991  3092.963  3085.374  3079.887
      [,9]      [,10]     [,11]     [,12]
model lm,13      lm,13      lm,13      lm,13
aic    1597.281  1594.994  1587.877  1576.677
bic    3077.512  3079.452  3076.561  3069.588

```

```
bic_values <- as.numeric(fit.forward$keep["bic", ])
```

```
bic_values <- na.omit(bic_values)
```

```

if (length(bic_values) > 0) {
  num_predictors <- seq_along(bic_values)

  plot(
    num_predictors, bic_values, type = "b",
    xlab = "Number of Predictors", ylab = "BIC",
    main = "BIC vs Number of Predictors (Forward Selection)"
  )

```

```

best_model_index <- which.min(bic_values)
best_model <- fit.forward$keep["model", best_model_index]

```

```

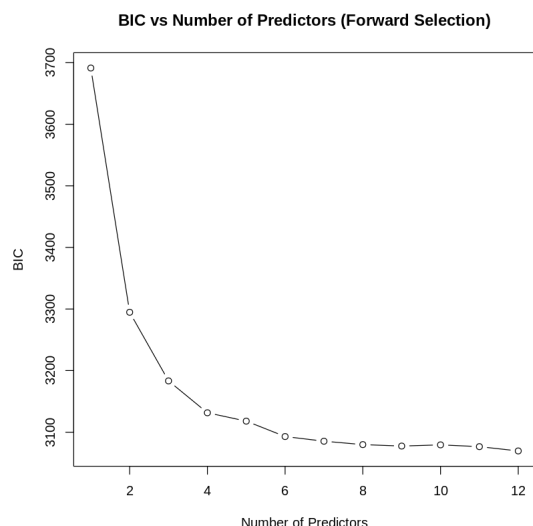
print(summary(best_model))
} else {
  print("No valid BIC values found.")
}

```

```

[1] Length Class Mode
model 13      lm      list

```



```

selected_model <- fit.forward$keep["model", stop_index][[1]]
print(summary(selected_model))

```

```

Call:
lm(formula = cmedv ~ lstat + rm + ptratio + dis + nox + chas +
    b + zn, data = dat)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-15.7094  -2.7787  -0.5822   1.5596  27.4931

```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  30.486661   4.827593   6.315 6.00e-10 ***
lstat        -0.549530   0.047229  -11.635 < 2e-16 ***
rm           4.087356   0.404965   10.093 < 2e-16 ***
ptratio      -0.864678   0.114690   -7.539 2.26e-13 ***
dis          -1.408006   0.185937   -7.572 1.80e-13 ***
nox          -16.951752   3.200194   -5.297 1.77e-07 ***
chas1         3.123190   0.862348    3.622 0.000323 ***
b             0.009418   0.002616    3.601 0.000349 ***
zn           0.038473   0.013180    2.919 0.003671 **
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 4.804 on 497 degrees of freedom
```

Multiple R-squared: 0.7306, Adjusted R-squared: 0.7262
F-statistic: 168.5 on 8 and 497 DF, p-value: < 2.2e-16

we select `lm(formula = cmedv ~ lstat + rm + ptratio + dis + nox + chas + b + zn, data = dat)` because it's last model before BIC value starts to increase.

```
fit.backward <- step(fit.full,
  direction = "backward",
  trace = FALSE,
  k = log(nrow(dat)),
  keep = function(model, aic) list(model = model, bic = BIC(model)))
```

```
bic_values <- as.numeric(fit.backward$keep["bic", ])
```

```
bic_values <- na.omit(bic_values)
```

```
if (length(bic_values) > 0) {
  stop_index <- which(diff(bic_values) > 0)[1]
```

```
  if (is.na(stop_index)) {
    stop_index <- length(bic_values)
  }
}
```

```
selected_model <- fit.backward$keep["model", stop_index][[1]]
```

```
print(summary(selected_model))
} else {
  print("No valid BIC values found.")
}
```



Call:
`lm(formula = cmedv ~ crim + zn + chas + nox + rm + dis + rad + tax + ptratio + b + lstat, data = dat)`

Residuals:

Min	1Q	Median	3Q	Max
-15.566	-2.686	-0.552	1.790	26.167

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	36.244827	5.022209	7.217	2.02e-12 ***
crim	-0.106657	0.032487	-3.283	0.001099 **
zn	0.047099	0.013402	3.514	0.000481 ***
chas1	2.727209	0.846606	3.221	0.001360 **
nox	-17.316823	3.503652	-4.943	1.06e-06 ***
rm	3.778662	0.402685	9.384	< 2e-16 ***
dis	-1.520270	0.184071	-8.259	1.35e-15 ***
rad	0.296555	0.062836	4.720	3.08e-06 ***
tax	-0.012077	0.003342	-3.613	0.000333 ***
ptratio	-0.917035	0.127912	-7.169	2.77e-12 ***
b	0.009202	0.002650	3.473	0.000561 ***
lstat	-0.528441	0.047001	-11.243	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.694 on 494 degrees of freedom
Multiple R-squared: 0.7444, Adjusted R-squared: 0.7387
F-statistic: 130.8 on 11 and 494 DF, p-value: < 2.2e-16

we select `lm(formula = cmedv ~ crim + zn + chas + nox + rm + dis + rad + tax + ptratio + b + lstat, data = dat)` as it is the last model before the BIC value starts to increase, meaning that removing more predictors would not improve our model.

```
fit.null <- lm(cmedv ~ 1, data = dat)
```

```
fit.full <- lm(cmedv ~ ., data = dat)
```

```
fit.stepwise.aic <- step(fit.null,
  scope = list(lower = fit.null, upper = fit.full),
  direction = "both",
  trace = FALSE)
```

```
summary(fit.stepwise.aic)
```



```
Call:
lm(formula = cmedv ~ lstat + rm + ptratio + dis + nox + chas +
  b + zn + crim + rad + tax, data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-15.566	-2.686	-0.552	1.790	26.167

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	36.244827	5.022209	7.217	2.02e-12 ***
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rm	3.778662	0.402685	9.384	< 2e-16 ***
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b	0.009202	0.002650	3.473	0.000561 ***
zn	0.047099	0.013402	3.514	0.000481 ***
crim	-0.106657	0.032487	-3.283	0.001099 **
rad	0.296555	0.062836	4.720	3.08e-06 ***
tax	-0.012077	0.003342	-3.613	0.000333 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.694 on 494 degrees of freedom
 Multiple R-squared: 0.7444, Adjusted R-squared: 0.7387
 F-statistic: 130.8 on 11 and 494 DF, p-value: < 2.2e-16

```
fit.stepwise.bic <- step(fit.null,
  scope = list(lower = fit.null, upper = fit.full),
  direction = "both",
  trace = FALSE,
  k = log(nrow(dat)))
```

```
summary(fit.stepwise.bic)
```



```
Call:
lm(formula = cmedv ~ lstat + rm + ptratio + dis + nox + chas +
  b + zn, data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-15.7094	-2.7787	-0.5822	1.5596	27.4931

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	30.486661	4.827593	6.315	6.00e-10 ***
lstat	-0.549530	0.047229	-11.635	< 2e-16 ***
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dis	-1.408006	0.185937	-7.572	1.80e-13 ***
nox	-16.951752	3.200194	-5.297	1.77e-07 ***
chas1	3.123190	0.862348	3.622	0.000323 ***
b	0.009418	0.002616	3.601	0.000349 ***
zn	0.038473	0.013180	2.919	0.003671 **

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 Multiple R-squared: 0.7306, Adjusted R-squared: 0.7262
 F-statistic: 168.5 on 8 and 497 DF, p-value: < 2.2e-16

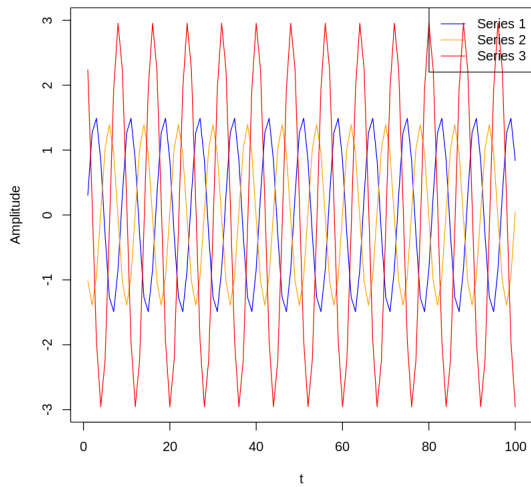
Q2, part d

```
omega <- pi / 4
n <- 100
set.seed(42)
```

```
simulate <- function(n, omega) {
  A <- rchisq(1, df = 1)
  Theta <- runif(1, 0, 2 * pi)
  t <- 1:n
  x_t <- A * cos(omega * t + Theta)
  return(x_t)
}
```

```
s1 <- simulate(n, omega)
s2 <- simulate(n, omega)
s3 <- simulate(n, omega)
```

```
plot(1:n, series1, type = "l", col = "blue", ylim = range(c(s1, s2, s3)),  
     ylab = "Amplitude", xlab = "t")  
lines(1:n, series2, col = "orange")  
lines(1:n, series3, col = "red")  
legend("topright", legend = c("Series 1", "Series 2", "Series 3"),  
      col = c("blue", "orange", "red"), lty = 1)
```



```
amplitude_series1 <- max(abs(s1))  
amplitude_series2 <- max(abs(s2))  
amplitude_series3 <- max(abs(s3))
```

```
amplitude_series1  
amplitude_series2  
amplitude_series3
```



```
1.49040723893573  
1.3928184858563  
2.95626337936125
```

Manideep Pendyala

problem 2.

A and Θ are independent random variables with $A \geq 0$ and $\Theta \sim \text{Unif}(0, 2\pi]$

$$x_t = A \cos(\omega t + \Theta), \quad t \in \mathbb{R}$$

(a)

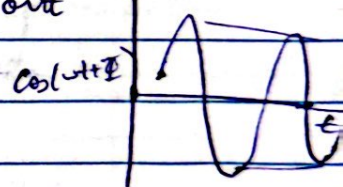
$$\mu_t = E x_t$$

$$= E[A \cos(\omega t + \Theta)]$$

$$= E(A) \cdot E[\cos(\omega t + \Theta)]$$

$$\mu_t = 0$$

Zero as this function is symmetric, so it gets canceled out



(b)

$$\gamma_{t, t+h} = \text{Cov}(x_t, x_{t+h})$$

$$= E[x_t x_{t+h}] - E[x_t] E[x_{t+h}]$$

$$= E[A \cos(\omega t + \Theta) A \cos(\omega t + \omega h + \Theta)]$$

$$= E[A \cos(\omega t + \Theta)] E[A \cos(\omega t + \omega h + \Theta)]$$

$$= E[A^2 \cos(\omega t + \Theta) \cos(\omega t + \omega h + \Theta)] - 0 \text{ from (a)}$$

$$= E[A^2] E[\cos(\omega t + \Theta) \cos(\omega t + \omega h + \Theta)]$$

$$A + B = \omega t + h\omega t + \phi$$

$$A - B = \omega t + \phi$$

$$2A = 2\omega t + h\omega t + 2\phi$$

$$2B = h\omega t$$

$$= E(A^2) E\left[\cos(\omega t + \phi) \cos(\omega t + h\omega t + \phi)\right]$$



$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$= E(A^2) E\left[\cos\left(\omega t + \frac{h\omega t}{2} + \phi\right) + \cos\left(\frac{h\omega t}{2}\right)\right] \frac{1}{2}$$

$$= \frac{1}{2} E(A^2) E[\cos \omega h] \quad (\because \omega \text{ is constant})$$

$$= \frac{1}{2} E(A^2) \cos h$$

③ $\rightarrow$ mean = 0, long term average of x_t is zero, means is symmetric distribution obvious from the cosine term

$\rightarrow$ Covariance has only one variable h ($h \in \mathbb{R}$), it's independent on that only, cosine makes it periodic based on value of ω (cosine has period 2π)
 $\cos(\pi + \theta) = -\cos \theta$

④ code, it's sinusoidal just like my plot in part ③

Amplitude is determined by value of A

problem 3

$$\bar{Y} = \frac{1}{n} \sum_{t=1}^n Y_t$$

① $\text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n} (Y_1 + \dots + Y_n)\right)$

$$= \frac{1}{n^2} \text{Var}(Y_1 + \dots + Y_n)$$

As Y_i 's are iid, and $\text{Var}(Y_i) = \sigma^2$

$$\text{Var}(\bar{Y}) = \frac{1}{n^2} (\sigma^2 + \dots + \sigma^2 \text{ --- } n \text{ times})$$

$$= \frac{1}{n^2} \cdot n \sigma^2 = \frac{\sigma^2}{n}$$

② Y_t can be a stationary function
its mean μ is similar to
1st order stationary,

$$\bar{Y} = \bar{X} = \frac{1}{T} \sum_{t=1}^T X_t$$

$$\text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n} \sum_{t=1}^n Y_t\right)$$

$$= \frac{1}{n^2} \text{Var}\left(\sum_{t=1}^n Y_t\right)$$

$$= \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}(Y_t, Y_s)$$

$$\text{w.k.T, } \gamma_{\Delta,t} = \text{Cov}(X_{\Delta}, X_t) \\ = E[(X_{\Delta} - \mu)(X_t - \mu)]$$

$$\text{Var}(\bar{Y}) = \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \gamma_{s,t}$$

$$k = |t-s|$$

$$s: 1 \rightarrow n$$

$$t: 1 \rightarrow n$$

$$\text{Var}(\bar{Y}) = \frac{1}{n^2} \left(n\gamma_0 + 2 \sum_{k=1}^{n-1} (n-k)\gamma_k \right)$$

we separate when they're equal

because we will have n elements in diagonal

each pair then γ_0 is $\text{Var}(Y_t)$ contributes twice

$$\text{Var}(\bar{Y}) = \frac{\gamma_0}{n} + \frac{2}{n} \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \gamma_k$$

③ non-stationary $\rightarrow Y_t = Y_{t-1} + \epsilon_t$

just like before

$$\text{Var}(\bar{Y}) = \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}(Y_t, Y_s)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(\epsilon_i, \epsilon_j)$$

$$\begin{aligned} & \sum_{i=1}^n \text{Var}(\epsilon_i) & \sum_{i=1}^n \text{Var}(\epsilon_i) \\ & \sum_{i=1}^n 1 & \sum_{i=1}^n 1 \end{aligned}$$

$$= n$$

$$s = \frac{1^2 + 2^2 + \dots + n^2}{n}$$

$$= \frac{1}{n^2} \sum_{k=1}^n (s_k +) \sigma^2$$

$$= \frac{1}{n^2} \sum_{k=1}^n k^2 \sigma^2$$

$$= \frac{1}{n^2} \cdot \sum_{k=1}^n k^2 \sigma^2 \quad \left(\begin{array}{l} \text{not sure} \\ \text{how, but} \\ \text{realised with} \end{array} \right.$$

$$= \frac{1}{n^2} \sigma^2 \cdot \frac{n(2n+1)(n+1)}{6} \quad \left(\begin{array}{l} \text{get answer} \\ \text{by substituting} \\ \text{hint} \end{array} \right)$$

$$= \frac{\sigma^2 (2n+1)(n+1)}{6n}$$